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UNIT 2
INORGANIC CHEMISTRY

Inorganic chemistry is the study of all the elements in the Periodic Table and the compounds they form, except organic compounds formed by carbon. The properties of these elements are very different and they form a huge variety of compounds. The Periodic Table is the great unifying principle used by inorganic chemists as a guide to understanding the behaviour of the elements and their compounds. Most of the elements in the Periodic Table are metals and these are some of the most important materials that we use in everyday life. However, scientists are always searching for new materials with exciting properties. Inorganic chemists are involved in the development of these new materials, for instance high-temperature superconductors that are used in trains that levitate above the tracks.



▲ Figure 11.1 The maglev train levitates above the track due to superconducting materials.

11 THE ALKALI METALS

We have already looked at the Periodic Table in Chapter 4. Here we will look at the properties of the elements in Group 1 of the Periodic Table: the alkali metals.



▶ Figure 11.2 Potassium reacting with water. The alkali metals are all reactive metals that react vigorously with water.

LEARNING OBJECTIVES

- Understand how the similarities in the reactions of these elements with water provide evidence for their recognition as a family of elements
- Understand how the differences between the reactions of these elements with air and water provide evidence for the trend in reactivity in Group 1
- Use knowledge of trends in Group 1 to predict the properties of other alkali metals

CHEMISTRY ONLY

- Explain the trend in reactivity in Group 1 in terms of electronic configurations.

Li	lithium
Na	sodium
K	potassium
Rb	rubidium
Cs	caesium
Fr	francium

▲ Figure 11.3 The alkali metals

The elements in Group 1 of the Periodic Table are called the **alkali metals**. The group contains the elements shown in Figure 11.3.

Francium (pronounced france-ee-um), at the bottom of the group, is **radioactive**. One of its isotopes is produced during the radioactive decay of uranium-235, but is extremely short-lived. At any one time scientists estimate that there is only about 20–30 g of francium present in the whole of the Earth's crust and no one has ever seen a piece of francium. When you know about the rest of Group 1 you can predict what francium would be like. We will make those predictions later.

PHYSICAL PROPERTIES

	Melting point/°C	Boiling point/°C	Density/g/cm^3
Li	181	1342	0.53
Na	98	883	0.97
K	63	760	0.86
Rb	39	686	1.53
Cs	29	669	1.88

EXTENSION WORK

The reason that the melting points decrease is that the atoms get bigger as we go down the group. In the metallic lattice, the nuclei of the positive ions are further from the delocalised electrons in caesium than in sodium and therefore there is weaker electrostatic attraction.

The melting and boiling points of the elements are very low for metals, and get lower as you move down the group.

Their densities tend to increase down the group, although not regularly. Lithium, sodium and potassium are all less dense than water, and so will float on it.

The metals are also very soft and are easily cut with a knife, becoming softer as you move down the group. They are shiny and silver when freshly cut, but tarnish very quickly on exposure to air.

STORAGE AND HANDLING



▲ Figure 11.4 Lithium, sodium and potassium have to be kept in oil to stop them reacting with oxygen in the air.

All these metals are extremely reactive and get more reactive as you go down the group. They all react quickly with oxygen in the air to form oxides, and react rapidly with water to form strongly alkaline solutions of the metal hydroxides. This is why the Group 1 metals are commonly known as the alkali metals.

To stop them reacting with oxygen or water vapour in the air, lithium, sodium and potassium are stored under oil. Rubidium and caesium are so reactive that they have to be stored in sealed glass tubes to stop any possibility of oxygen getting at them.

Great care must be taken not to touch any of these metals with bare fingers. There could be enough sweat on your skin to give a reaction, producing lots of heat and a very **corrosive** metal hydroxide.

A FAMILY OF ELEMENTS

DID YOU KNOW?

Historically alkali metals were put in the same group because of similarities in properties, before anyone knew anything about electrons!

There are two reasons that we put these elements in Group 1:

1 They all have *one* electron in their outer shell. The electronic configurations are:

lithium	2, 1
sodium	2, 8, 1
potassium	2, 8, 8, 1

2 They have similar chemical properties, for example:

- they all react with water (this will be discussed below) in the same way to form a hydroxide with the formula MOH (e.g. LiOH, NaOH) and hydrogen
- they react with oxygen to form an oxide with the formula M_2O (Na_2O, K_2O)
- they react with halogens to form compounds with the formula MX (e.g. LiCl, KBr)
- they form ionic compounds which contain an M^+ ion (e.g. Na^+, K^+).

KEY POINT

'M' simply represents any one of the alkali metals. 'X' represents any one of the halogens.

The chemical properties depend on the number of electrons in the outer shell. The Group 1 elements react in very similar ways because they all have the same number of electrons in the outer shell (one), so reason 2 is really just a consequence of reason 1.

We will discuss the chemical properties in more detail below.

REACTIONS WITH WATER

All these metals react in the same way with water to produce a metal hydroxide and hydrogen:

alkali metal + water → alkali metal hydroxide + hydrogen

$2M + 2H_2O \rightarrow 2MOH + H_2$

The main difference between the reactions is how quickly they happen.

As you go down the group, the metals become more reactive and the reactions occur more rapidly.

The reaction between sodium and water is typical.

REMINDER

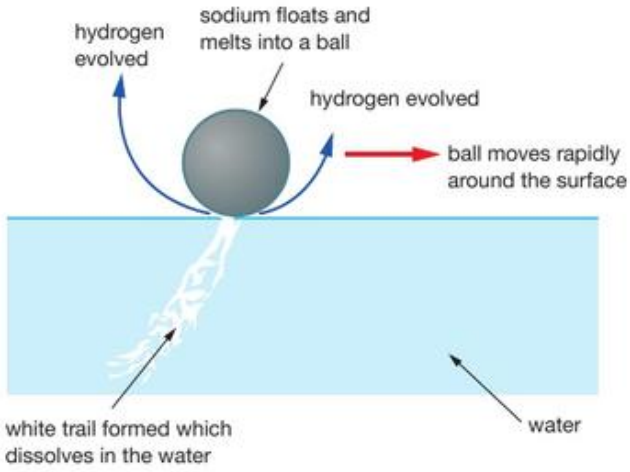
You might need to remind yourself why the formula is MOH by looking at Chapter 7. The charge on the hydroxide ion is 1–.

WITH SODIUM

$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

HINT

Strictly speaking, most of the time the sodium is reacting it is present as molten sodium, not solid sodium. Writing (l) for the state symbol has the potential to confuse an examiner and is probably best avoided!



▲ Figure 11.5 Sodium reacting with water. The white trail is the sodium hydroxide, which dissolves in water to form a strongly alkaline solution.

The main observations you can make when this reaction occurs are:

- The sodium floats because it is less dense than water.
- The sodium melts into a ball because its melting point is low and a lot of heat is produced by the reaction.
- There is fizzing because hydrogen gas is produced.
- The sodium moves around on the surface of the water. Because the hydrogen isn't given off symmetrically around the ball, the sodium is pushed around the surface of the water, like a hovercraft.
- The piece of sodium gets smaller and eventually disappears. The sodium is used up in the reaction.
- If you test the solution that is formed with universal indicator solution, you will see that the universal indicator goes blue, indicating an alkaline solution has been formed. The metal hydroxide is alkaline (the solution contains the OH^- ion).

HINT

When asked to write *observations* in the exam it is better to write 'fizzing/bubbling/effervescing' rather than 'a gas is given off' because the fizzing/bubbling/effervescing is what you actually see.



▲ Figure 11.6 **A hovercraft**

HINT

When you are asked about this in the exam, you are often asked to compare the reactions of sodium and lithium so you should explain how you can see that the reaction of lithium is slower. So, for example, you can say that it fizzes more slowly, or the lithium moves around more slowly, or takes longer to disappear.

HINT

Again, if you are asked to compare potassium with sodium use phrases such as:
'fizzes *more* vigorously'
'moves around *more* quickly'
'disappears *more* quickly'
The key difference though is that with potassium the hydrogen bursts into flames but with sodium it usually does not.

!

Safety Note: The reactions of rubidium and caesium would be too hazardous to attempt in school as they would explode.

LITHIUM

$$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$$

The reaction is very similar to sodium's reaction, except that it is slower. Lithium's melting point is higher and the heat isn't produced so quickly, so the lithium doesn't melt.

POTASSIUM

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

Potassium's reaction is faster than sodium's. Enough heat is produced to ignite the hydrogen, which burns with a lilac flame. The reaction often ends with the potassium spitting around and exploding.

RUBIDIUM AND CAESIUM

These react even more violently than potassium, and the reaction can be explosive. Rubidium hydroxide and caesium hydroxide are formed.

CHEMISTRY ONLY

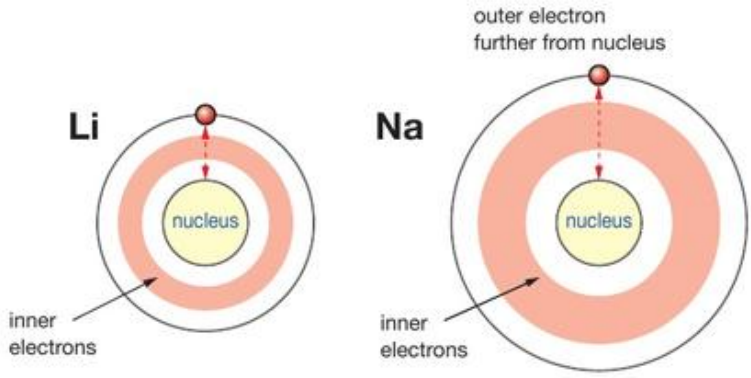
EXPLAINING THE INCREASE IN REACTIVITY

As you go down the group, the metals become more reactive.

In all these reactions, the metal atoms are losing electrons and forming metal ions in solution. For example:

$$Na(s) \rightarrow Na^+(aq) + e^-$$

The differences between the reactions depend on how easily the *outer electron* of the metal is lost in each case. That depends on how strongly it is attracted to the nucleus in the original atom. Remember that the nucleus of an atom is positive because it contains protons, and so attracts the negative electrons.



▲ Figure 11.7 Electrons of lithium and sodium

EXTENSION WORK

The electrons released by the metal are gained by the water molecules, producing hydroxide ions and hydrogen gas.

$2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$

As we move down the group, the atoms have more shells of electrons and get bigger: a sodium atom is bigger than a lithium atom and a potassium atom is bigger than a sodium atom. As the atoms get bigger, the *outer electron*, which is the one lost in the reaction, is further from the nucleus. Because it is further from the nucleus it is less strongly attracted by the nucleus and therefore more easily lost.

END OF CHEMISTRY ONLY



▲ Figure 11.8 A piece of sodium. The left-hand edge has been freshly cut, so it is shiny.

REACTIONS OF THE ALKALI METALS WITH THE AIR

Lithium, sodium and potassium are all stored in oil because they react with the air. If we look at a piece of sodium which has been taken out of the oil, it usually has a crust on the outside. It is not shiny unless it has been freshly cut.

When the piece of sodium is cut, the fresh surface is shiny but it tarnishes rapidly as the freshly exposed sodium reacts with oxygen in the air. If we do the same with a piece of lithium it tarnishes more slowly because lithium reacts more slowly than sodium. A freshly cut piece of potassium tarnishes extremely rapidly, more quickly than sodium. In this way we can see again that potassium is more reactive than sodium, which is more reactive than lithium. In each case the metal reacts with oxygen in the air to form an oxide with the formula M_2O.

If we heat each of the metals in the air using a Bunsen burner, we get a much more vigorous reaction and it is more difficult to see which metal is most reactive because all the reactions are so rapid.

Lithium burns with a red flame to form lithium oxide.

Sodium burns with a yellow flame to form sodium oxide.

Potassium burns with a lilac flame to form potassium oxide.

The equation for all these reactions is:

$$4M(s) + O_2(g) \rightarrow 2M_2O(s)$$

In each case the product formed is a white powder – the alkali metal oxide.

REMINDER

Remember the charge on the oxide ion is O^{2-} and the charge on an alkali metal ion is M^+.

EXTENSION WORK

When the alkali metals react with air, other types of oxide can also be formed, called peroxides (M_2O_2) and superoxides (MO_2). These all still contain the M^+ ion, it is the oxygen bit which is different. Lithium can also form a nitride (Li_3N).

COMPOUNDS OF THE ALKALI METALS

All Group 1 metal ions are colourless. That means that their compounds will be colourless or white unless they are combined with a coloured negative ion. Potassium dichromate(VI) is orange, for example, because the dichromate(VI) ion is orange, and potassium manganate(VII) is purple because the manganate(VII) ion is purple. Group 1 compounds are typical ionic solids and are mostly soluble in water.

SUMMARISING THE MAIN FEATURES OF THE GROUP 1 ELEMENTS

Group 1 elements:

- are metals
- are soft with melting points and densities which are very low for metals
- have to be stored out of contact with air or water
- react rapidly with air to form coatings of the metal oxide
- react with water to produce an alkaline solution of the metal hydroxide and hydrogen gas
- increase in reactivity as you go down the group
- form compounds in which the metal has a 1+ ion
- have mainly white/colourless compounds which dissolve to produce colourless solutions.

PREDICTING THE PROPERTIES OF FRANCIUM

As we move down a group in the Periodic Table the properties of the elements change gradually. So, if we know the properties of most of the elements in a group, we should be able to predict the properties of elements we don't know. Francium is extremely radioactive and at any time, anywhere in the world, there is only a tiny amount present; nobody has actually seen a piece of francium. We can, however, predict the properties of francium using the properties of the other alkali metals.

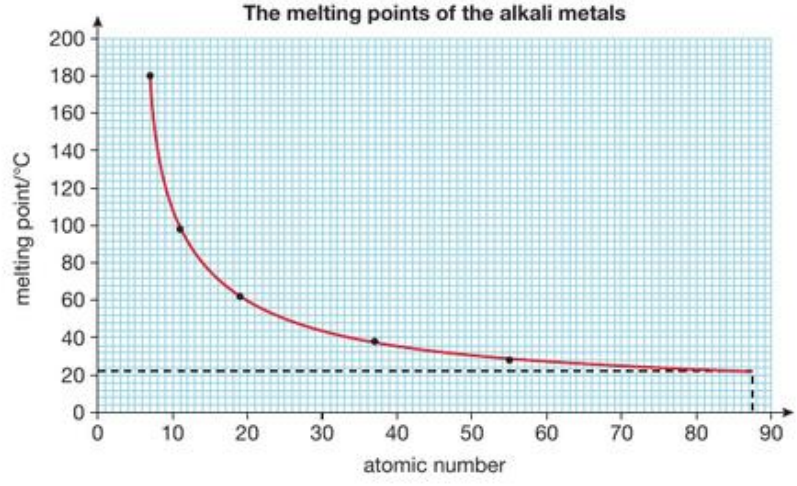
We can predict that francium:

- is very soft
- will have a melting point around room temperature
- has density which is probably just over 2 g/cm^3
- will be a silvery metal, but will tarnish almost instantly in air
- will react violently with water to give francium hydroxide and hydrogen
- will be more reactive than caesium
- will have a hydroxide, francium hydroxide, with the formula FrOH, which will be soluble in water and form a strongly alkaline solution
- will form compounds that are white/colourless and dissolve in water to give colourless solutions.

REMINDER

Remember, you can't actually observe any of these things.

We could use a graphical method to predict the melting point of francium. If we plot the melting point of the alkali metals against atomic number then draw a line of best fit we get:



▲ Figure 11.9 This graph allows us to predict the melting point of francium.

KEY POINT

Various other predictions give a melting point for francium between 21 °C and 27 °C.

If we carry on the line to atomic number 87 we can predict a melting point of about 22 °C.

CHAPTER QUESTIONS

SKILLS REASONING 7

1 Explain why lithium, sodium and potassium are put in the same group in the Periodic Table.

SKILLS CRITICAL THINKING

2 This question concerns the chemistry of the elements Li, Na, K, Rb and Cs. In a and b, you should name the substances represented by letters.

a **A** is the least dense of all the metals.

b When metal **B** is dropped onto water it melts into a small ball and moves rapidly around the surface. A gas, **C**, is given off and this burns with a lilac flame. A solution of **D** is formed.

9 c Write an equation for the reaction of **B** with water.

SKILLS REASONING 6 d What would you expect to see if solution **D** was tested with universal indicator paper?

7 e Explain why **B** melts into a small ball when it is dropped onto water.

SKILLS CRITICAL THINKING 9 f **E** burns in air with a yellow flame to form compound **F**. Write a word equation and balanced symbol equation for the reaction that occurs.

7 3 Explain whether each of the following statements is true or false.

a Sodium forms mostly covalent compounds.

8 b A rubidium atom is larger than a potassium atom.

6 c All the alkali metals react with air to form oxides.

7 d Lithium reacts with chlorine to form lithium chloride, which has the formula Li_2Cl.

SKILLS REASONING 8 4 Imagine that a new alkali metal has recently been discovered and that it fits into the Periodic Table below francium. We will call this new element edexcelium.

a Explain whether you would expect edexcelium to be more or less dense than francium.

SKILLS PROBLEM SOLVING 9 b State how many electrons edexcelium will have in its outer shell.

SKILLS CRITICAL THINKING 8 c State the names of the products that will be formed when edexcelium reacts with water.

d Explain whether edexcelium will be more or less reactive than francium.

SKILLS PROBLEM SOLVING 9 e If the symbol for edexcelium is Ed, write a balanced chemical equation for the reaction of edexcelium with water.

SKILLS CRITICAL THINKING 6 f When edexcelium reacts with water, will the solution formed be acidic, alkaline or neutral?

SKILLS PROBLEM SOLVING 7 g Write the formula for the compound formed when edexcelium reacts with air.